

Business Task Summary

Analyze smart device usage data from non-Bellabeat users to identify trends in how consumers use wearable fitness devices. Apply these insights to one Bellabeat product and provide recommendations that can help inform Bellabeat's marketing strategy and support future business growth.

Stake holders

Urška Sršen – Bellabeat co-founder and Chief Creative Officer.

Sando Mur – Bellabeat co-founder and key member of the executive team.

Bellabeat Marketing Analytics Team – Responsible for collecting, analyzing, and reporting data to guide Bellabeat's marketing strategy.

Questions to answer

What trends can be identified in smart device usage among consumers?

How can these trends be applied to Bellabeat customers and products?

How can these insights be used to influence Bellabeat's marketing strategy and drive future growth?

Description of data sources

FitBit Fitness Tracker Data (CC0: Public Domain, data set made available through Mobius): This Kaggle dataset contains personal fitness tracker data from thirty fitbit users. Thirty eligible Fitbit users consented to the submission of personal tracker data, including minute-level output for physical activity, heart rate, and sleep monitoring. It includes information about daily activity, steps, and heart rate that can be used to explore users' habits.

The dataset was imported into google Big Query for preparation and analysis

The dataset Daily Activity includes 940 total records at a daily user level (one record per day).

- **Id:** Unique identifier for each Fitbit user
- **ActivityDate:** The date of the recorded activity
- **TotalSteps:** Total number of steps taken in a day
- **TotalDistance:** Total distance travelled in a day
- **TrackerDistance:** Distance tracked by the Fitbit device
- **LoggedActivitiesDistance:** Distance from manually logged activities
- **VeryActiveDistance:** Distance covered in very active activity
- **ModeratelyActiveDistance:** Distance covered in moderate activity
- **LightActiveDistance:** Distance covered in light activity
- **SedentaryActiveDistance:** Distance recorded during sedentary activity
- **VeryActiveMinutes:** Minutes spent in very active activity
- **FairlyActiveMinutes:** Minutes spent in moderately active activity
- **LightlyActiveMinutes:** Minutes spent in light activity
- **SedentaryMinutes:** Minutes spent sedentary
- **Calories:** Estimated calories burned per day

The dataset provides a combination of distance-based, time-based, and energy expenditure metrics, allowing analysis of user activity patterns, physical intensity levels, and overall daily behaviour.

Documentation of Cleaning or Manipulation to Data

The dataset was imported into Google BigQuery and underwent several validation and quality checks prior to analysis.

1. Duplicate check

A query was ran to identify any duplicates of the data. No duplicates were found, confirming that each user had unique entries.

2. Missing Value check

A query was ran to identify any NULL values in the data. No missing values were detected in the dataset.

3. Time range validation

A query was ran to confirm that the time range matched the initial description of the data timeframe which was 31 days. The query confirmed that the data did in fact lie within the specified range.

4. User participation consistency check

User logged completeness was evaluated by running a query which counted the number of active tracking days per user. While most users had close to 30–31 days of data, some users had reduced participation (as low as 4 days). This shows uneven user engagement or incomplete tracking across the dataset. Additionally, a daily coverage check showed that the number of active users varied across the time period, with a general decline in participation toward the end of the dataset period.

5. Data filtering

None of the data was removed or filtered.

Analysis

The analysis focused on three aspects of smart device usage that are relevant to Bellabeat's business objectives: activity intensity, differences in user activity levels, and user engagement over time.

Activity Intensity Distribution

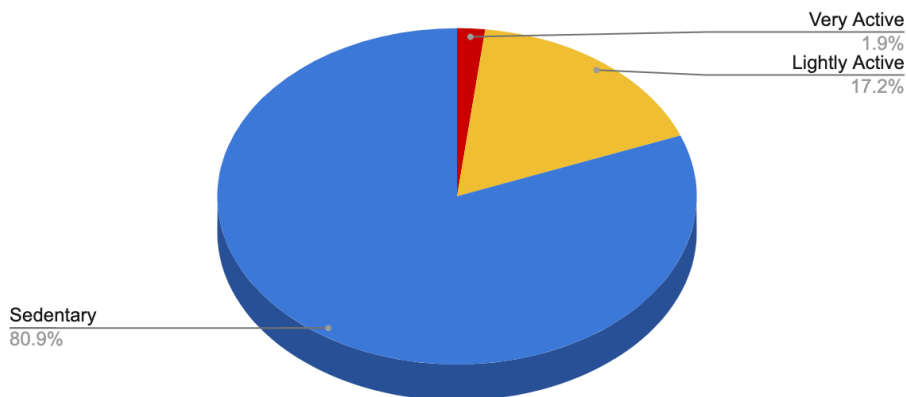
I wanted to find out what was the average daily percentage for each active state level; Very active, lightly active and sedentary.

I ran the following query to find the average and divide it into the three groups:

```

Daily Activity Type Percentages
Run Save Schedule Share
File Edit Run Tools
1 SELECT
2   ROUND(AVG(100 * SAFE_DIVIDE(VeryActiveMinutes,
3     VeryActiveMinutes + LightlyActiveMinutes + SedentaryMinutes)), 2) AS avg_pct_very_active,
4
5   ROUND(AVG(100 * SAFE_DIVIDE(LightlyActiveMinutes,
6     VeryActiveMinutes + LightlyActiveMinutes + SedentaryMinutes)), 2) AS avg_pct_lightly_active,
7
8   ROUND(AVG(100 * SAFE_DIVIDE(SedentaryMinutes,
9     VeryActiveMinutes + LightlyActiveMinutes + SedentaryMinutes)), 2) AS avg_pct_sedentary
10 FROM couresa-497016.Project.daily_activity
11
12
```

Activity Intensity Distribution



The results show the average Fitbit user in this dataset spends about 81% of their tracked time sedentary, 17% lightly active, and less than 2% very active. This indicates that intense physical activity represented only a small proportion of users' recorded daily activity.

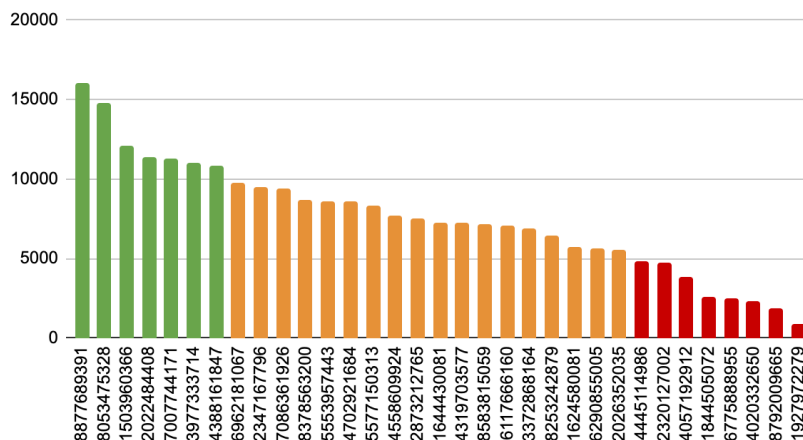
User Activity Segmentation

I wanted to understand how users differed in their activity levels. I decided the daily step count is a good indicator for this.

I ran a query dividing the participants into 3 groups; low, moderate and high activity levels in order to find how much the group varied.

```
Percentage of Activity Level [Run] [Save] [Schedule] [Share]
File Edit Run Tools
1 SELECT
2   activity_level,
3   COUNT(*) AS users
4 FROM (
5   SELECT
6     Id,
7     CASE
8       WHEN AVG(TotalSteps) < 5000 THEN 'Low'
9       WHEN AVG(TotalSteps) < 10000 THEN 'Moderate'
10      ELSE 'High'
11    END AS activity_level
12 FROM `couresa-497016.Project.daily_activity`
13 GROUP BY Id
14 )
15 GROUP BY activity_level;
```

Users Activity Level By Steps



Analysis of user activity levels showed behavioural segments within the Fitbit user base. Over half of users (54.55%) fall into the moderate activity category, while only 21.21% can be classified as highly active ($\geq 10,000$ steps per day). A further 24.24% of users are in the low activity category ($< 5,000$ steps per day).

This indicates considerable variation in activity levels among Fitbit users, with only a small proportion consistently achieving high daily step counts

3. User Engagement Trends

I wanted to know if users were consistent with logging their days. I ran a query to find users engagement (amount of days logged) over the course of the month:

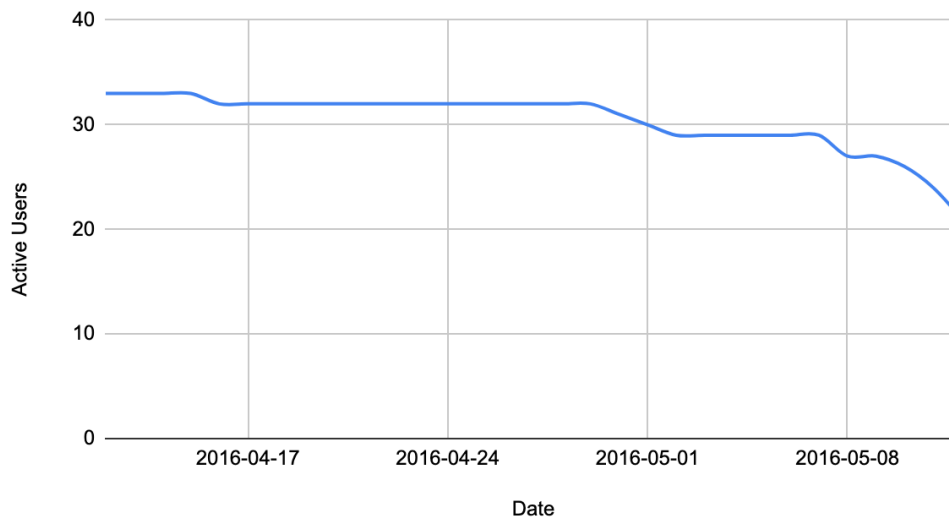
🔍 User Engagement

Run Save Schedule Share

File Edit Run Tools

```
1 SELECT
2   ActivityDate,
3   COUNT(DISTINCT Id) AS active_users
4 FROM `couresa-497016.Project.daily_activity`
5 GROUP BY ActivityDate
6 ORDER BY ActivityDate;
```

30 Day User Engagement



Daily activity logging declined from 33 users at the beginning of the study to 21 by the end, representing a 36.36% decrease in user participation over a 31 day period.

This suggests that user participation decreased throughout the 31-day study period.

Recommendations

Upon analysis we found that the average daily percentage for each active state level differs greatly, with users spending the vast amount of time sedentary. This finding suggests Bellabeat has an opportunity to encourage healthier daily habits among users.

Bellabeat should incorporate an avatar, a personal coach in their app to send reminders, provide recommendations and encourage users towards becoming healthier.

For the marketing team this adds a completely new dimension to what Bellabeat can offer.

Bellabeat is a tech company that designs beautiful smart devices for women to wear, with the goal of making women healthier through a personalized coach. By incorporating an avatar it adds another dimension of personalisation and helps keep accountability in a positive manner. If Bellabeat could encourage and instill healthier habits in women, such as being more active, it will be extremely positive for the brand.

By segmenting our users by their daily average step count we found that activity levels (by step count) vary considerably between Fitbit users every day. It also highlighted daily step count as a metric where most users have room for improvement.

When someone embarks on their health journey it can be an extremely confusing and intimidating place to be. Most people don't know what they need to be doing, how much of it and how many times per week to start becoming healthier. The step counter is a simple metric that everyone can understand and is a feature I suggest that Bellabeat leans on.

Since nearly 79% of users in the dataset were classified as low to moderately active, Bellabeat should make personalised step goals the first metric new users focus on. The step counter is an accessible, easy to understand starting point for beginners looking to improve their health. The Bellabeat app can show stats, visualisations and progress trackers to keep beginner users interacting and engaged. By emphasising the step counter it gives the marketing team an opportunity to target those who are new to fitness and show that the fitness journey is accessible to anyone.

We discovered that over the course of the month daily activity logging declined. We can't be certain as to exactly why Fitbit users' logging declined over time but we can be sure that user retention should be a priority for Bellabeat. The app could have a few unique, fun ways for users to see their exercise and health data. Potentially something similar to that of Spotify's Wrapped yearly posts. Bellabeat already has a unique angle that would work well with this idea as they have multiple products which track data that most other smart devices don't, such as menstrual cycles, water intake, sleep just to name a few. Monthly and annual personalised summaries could also encourage users to share their achievements on social media, increasing organic brand exposure and attracting new customers.

This would help Bellabeat by increasing brand awareness while reinforcing long-term engagement.

Conclusion

This analysis identified three key trends in Fitbit smart device usage: users spend the majority of their tracked time sedentary, activity levels vary considerably between individuals, and user engagement declines over time. By marketing accessible features such as personalised step goals and long-term health guidance, Bellabeat can appeal to the large population of people beginning their health and fitness journey. Features that encourage ongoing engagement and social sharing can also increase organic brand exposure, strengthen customer retention, and unlock new opportunities for growth by attracting and retaining a broader customer base.